

OS ASSIGNMENT 1

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Div & batch: B / B3

PS:Write a shell script to display disk space usage of the file system, including the total space, used space, available space, and usage percentage.

Code:

```
echo "-----"
echo "   DISK USAGE REPORT   "
echo "-----"

disk=$(df / | tail -1)

total=$(echo "disk" | awk '{print $2}')
used=$(echo "disk" | awk '{print $3}')
available=$(echo "disk" | awk '{print $4}')
percent=$(echo "disk" | awk '{print $5}')

total_gb=$((total/1024/1024))
used_gb=$((used/1024/1024))
available_gb=$((available/1024/1024))

echo "Total Space :${total_gb} GB"
echo "Used Sapce :${used_gb} GB"
echo "Avaliable :${available_gb} Gb"
echo "Dish Usage :${percent} "

echo "-----"

usage=${percent%%}%

if [ "$usage" -ge 80 ];
then
    echo "WARNING: HIGH DISK USAGE!!!"
else
    echo "Normal Disk Usage."
fi
```

```
echo "-----"
```

```
df -h --output=source,size,used,avail,pcent,target
```

```
echo "-----"
```

Output:

```
omion@OM: ~  
omion@OM:~$ ./disk_usage.sh  
-----  
DISK USAGE REPORT  
-----  
Total Space :1006 GB  
Used Sapce :2 GB  
Avaliable :953 Gb  
Dish Usage :1%  
-----  
Normal Disk Usage.  
-----  
Filesystem      Size  Used Avail Use% Mounted on  
none            7.8G   0 7.8G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2  
none            7.8G 4.0K 7.8G   1% /mnt/wsl  
drivers         149G 135G  14G  91% /usr/lib/wsl/drivers  
/dev/sdd        1007G 2.1G 954G   1% /  
none            7.8G 40K 7.8G   1% /mnt/wslg  
none            7.8G   0 7.8G   0% /usr/lib/wsl/lib  
rootfs          7.8G 2.8M 7.8G   1% /init  
none            7.8G 500K 7.8G   1% /run  
none            7.8G   0 7.8G   0% /run/lock  
none            7.8G   0 7.8G   0% /run/shm  
none            7.8G  80K 7.8G   1% /mnt/wslg/versions.txt  
none            7.8G  80K 7.8G   1% /mnt/wslg/doc  
C:\             149G 135G  14G  91% /mnt/c  
D:\             327G  37G 290G  12% /mnt/d  
none            1.0M   0 1.0M   0% /run/credentials/systemd-journald.service  
tmpfs           7.8G   0 7.8G   0% /tmp  
none            1.0M   0 1.0M   0% /run/credentials/systemd-resolved.service  
none            1.0M   0 1.0M   0% /run/credentials/getty@tty1.service  
tmpfs           1.6G  12K 1.6G   1% /run/user/1000  
-----  
omion@OM:~$ |
```

- positive or negative number
sys>> pos.sh

```
read choice
```

```
if [ $choice -gt 0 ] ; then
```

```
    echo "$choice number is positive"
```

```
else
```

```
    echo "$choice number is negative"
```

```
fi.
```

- arithmetic operation
sys>> arithmetic.sh

```
read a
```

```
echo "enter second number"
```

```
read b
```

```
echo "choice"
```

```
read choice
```

```
case $op in
```

```
  +) echo "result = $ ((a+b))" ;;
```

```
  -) echo "result = $ ((a-b))" ;;
```

```
  *) echo "result = $ ((a+b))" ;;
```

```
  /) echo "result = $ ((a/b))" ;;
```

```
  *) echo "Invalid" ;;
```

```
esac
```

→ loop

```
for i in {1...10}
```

```
do echo $i
```

```
done.
```

8) clear → erase all

9) man → gives all manuals of all commands

ex: mkdir om → makes directory named om

cd om → enters into om directory

ls → list all possible

cd .. → Exiting from working directory

pwd → present working directory

touch omx → creating file "omx"

rm omx → remove "omx" file

- To check which shell we are using
sys >> cat /etc/shells

- To create script file
sys >> vi myscript.sh

- To run a script
sys >> chmod +x filename/myscript.sh
sys >> ./myscript.sh

- if else statement syntax :

if [condition] ; then

echo " "

else

echo " "

else :

fi

Fi

- Print number from 1 to 10

! /bin/bash

i = 1

while [\$i -le 10]

do

echo \$i

i = \$(i+1)

done

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OS lab

JOHS	Page No.
Date	/ /

Commands :

Format : Command

Description

Syntax

Output

date

echo

who

whoami / who am i

tty

bc

clear

man

cal

mkdir cd cat

ls mv

mkdir pwd

Copy/cp

chmod

ntfs file system in windows
↳ to see this system information

- Commands, Description, Syntax & Output.

① Ls with option.

Cmd: List directory: List directory contents

Descrip: List information about files (in present directory). Sorted

Syntax: --> Ls [option] [file]

Out: Lists all present directories.

Ls at root = root & home Student . . .

to execute file : vi filename.sh def permission : read, write

\$ chmod +x yourfilename

\$ chmod +s filename

run : \$ bash bar/filename

\$ sh filename

\$./filename.

Basic Linux Commands

① File & Directory Management

- a) `cd` → change directory or change path.
change working directory by providing certain path to it
`cd ~/Users/OM/Downloads`

Attributes:

- l : force symbolic links to the
- p : physical directory structure

- b) `Cp` → copy file or directory command
used to copy files and directory from one location to another in linux and unix kind of systems

Attributes:

- r : Recursive (Required when copying directory)
- i : Interactive (Prompts you for confirmation before overwriting)
- u : Update (Copies the file only if the source is newer)

- c) `vi`, `vim`, `nano` → text editor (to create and edit file).

- d) `Cat` → Concatenate / Join files together

Attributes:

- t -n : Used to read script or configure file

- e) `mkdir` → to create folder / directory.

Attributes:

- p : Parent (most useful flag, create parent directories)
- v : Verbose (print message for every directory it successfully)

Comment: #

Linux variable $\rightarrow$ System variable

Capital case

$\rightarrow$ User defined variable

lower case

variable name = value eg: \$n=10

\$variable name.

~~echo~~ echo \$vch

eg: \$n=10

echo \$n

Arithmetic operations

\$ expr op1 operator op2 eg expr 1 + 3

Input: read name/ask to call variable put \$!.

Save program ^{save.} escape : \q \quit

f) `rm` → remove (used to delete files and directory)

Attributes

- i : Interactive (prompts you for confirmation b4 deleting files)
- r : Recursive (Required to delete directory)
- f : Force (Delete file without asking for confirmation)
- v : verbose (Prints the name of each file as it del successfully)

g) `mv` → Used to move file directory or rename any file

h) `pwd` → current file path

i) `ls` → listing information of file

j) `find` → finding files

k) `touch` → create file in current directory.

② Basic Other Commands

1) `date` → dates printing year (def: current)

2) `cal` → calendar (def: current)

3) `echo` → print some input

4) `who` → all users who logged in

5) `whoami` → currently who is logged in

6) `Tty` → terminal devices name

7) `bc` → basic calculations.

② echo

Display a line of text

Descrpt: Display the line of text provided to it

Syntax: --> echo [short-option]... [string] / echo [long-option]

Output --> echo "TY B div"

TY B div.

options: \a alert

\b backspace

\c escape

\n newline

③ who

Display who logged on.

--> who [option] [file] ARG1 ARG2

url:

whoami

to know my

Login Information

④ Cat

Concatenate files and print on the standard output.

--> Cat [option] [file]

Notebook: